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OSU AI REU – Final Report

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Introduction

The field of artificial intelligence has undergone significant developments over the past decade, with breakthroughs driven by the advent of machine learning, deep learning, and large-scale generative models. This paper reviews my two-month Research Experience for Undergraduates (REU) project, which provided a comprehensive introduction to modern AI systems, machine learning principles, and cutting-edge applications. In addition to summarizing the key topics covered during the project, this paper critically examines the paper “Mastering Go with Deep Neural Networks and Tree Search” by Silver et al. and provides a personal reflection on the implications and innovations outlined in the paper.

REU Learning Experience

During the two-month REU program, I was introduced to a broad spectrum of foundational and advanced topics in artificial intelligence and machine learning. The curriculum was designed to scaffold conceptual understanding and practical skills, culminating in critical engagement with recent developments in AI research. Below is a thematic summary of the topics covered.

Introduction to AI and ML

The program begins with an overview of AI and its historical development, then focuses on machine learning (ML), including supervised learning, unsupervised learning, and reinforcement learning paradigms. This lays the foundation for understanding algorithmic decision making and data-driven modeling.

Core ML Concepts

The core Machine Learning topics we cover includes regression, classification, clustering, and gradient descent. Through all the assignments and visual demonstrations, I was able to gain a deeper understanding of the processes involved in model training and evaluation. Additionally, I learned the importance of recognizing and addressing overfitting in models, especially through adjusting various parameters and observing how those changes affected the model's performance. This process helped me understand how tuning can improve generalization.

Autoregressive Transformers and Language Models

This project explored the architecture and mechanics of Autoregressive Transformers (similar to GPT models) which rely on self-attention mechanisms to capture contextual relationships in sequence data. I learned how these models can be trained on large-scale corpora to generate human-like text.

Wireless Networks and Probability

We also studied the basics of wireless network architecture and channel modeling, followed by probability theory fundamentals, including conditional probabilities

and Bayes' Rule. This mathematical foundation was essential for understanding statistical learning techniques.

Generative Models for Images

This was one of the areas that had the biggest impact on me. I worked on variational autoencoders (VAEs), generative adversarial networks (GANs), and diffusion models. Hands-on demos showed how these models can generate synthetic and realistic images. The interplay between loss functions, discriminator-generator dynamics, and latent space manipulation was particularly fascinating.

Generative Models for Text

Building on the transformer section, this part focused on models capable of producing coherent and stylistically consistent text. Topics included masked language modeling (as in BERT), decoder-only models (like GPT), and the role of fine-tuning for domain-specific tasks. As part of the coursework, I completed an assignment to train a small GPT model on a dataset of Shakespearean character dialogues, which helped me understand how these models learn and generate context-appropriate language.

Overall, the most intellectually rewarding topics were machine learning fundamentals and generative modeling techniques for both images and text. These topics not only captured the mathematical beauty of learning systems but also showcased their transformative power in creative and cognitive tasks.

Summary of the Paper: "Mastering the Game of Go with Deep Neural Networks and Tree Search"

The paper published in Nature in 2016 by Silver et al., discussed the design and training methods of AlphaGo, the first computer program that beat a professional player in an entire game of Go. This work is an important milestone in artificial intelligence research, demonstrating the successful combination of deep learning and tree search techniques.

Problem Setting

Go is a classic board game known for its perfect information, but its vast number of possible board configurations (about 10^{170}) makes traditional brute-force searching impractical. Previous attempts to create a computer program to master Go struggled due to the game's high branching factor and the absence of a reliable way to evaluate board positions.

Main Components of AlphaGo

The author summarizes AlphaGo as a combination of three key components. The first one is Policy Networks, it was trained using supervised learning (SL) and reinforcement learning (RL) to estimate the probability of legal moves at any board state. Second one is Value Networks, these networks estimate the expected outcome of the game based on a given board position. The last one is Monte Carlo Tree Search (MCTS), This approach allows for effective lookahead by simulating possible moves and selecting those that maximize expected value. To train the SL policy, the team used 30 million positions from expert games to predict the next move. The RL policy then improved upon this through self-play, focusing on maximizing winning chances rather

than just mimicking human strategies. The value network was trained to predict game outcomes of these self-play games.

Integration with MCTS

AlphaGo combines the policy and value networks within the MCTS framework to enhance exploration and board position evaluation. The search tree utilizes statistics like visit counts and action values, while the value network offers a more efficient alternative to traditional rollouts. The final move is determined by the highest visit count.

Performance and Evaluation

AlphaGo achieved a 5-0 victory against the European champion Fan Hui and boasted a remarkable 99.8% win rate against other Go programs. A distributed version of AlphaGo, which utilized more CPUs and GPUs, outperformed even the original single-machine setup.

Personal Understanding and Reflections on the Paper

AlphaGo embodies many of the ideas I explored during my REU, especially in generative models. What interested me most was its approach, rather than hard-code the algorithm, AlphaGo learned from data through professional player and self-play, marking a key shift toward end-to-end learning in AI. Its use of deep convolutional networks parallels my work in image generation models, where learning compact representations mirrors the principles of VAEs and GANs.

The paper's multi-stage training approach also caught my attention. The authors initially bootstrapped learning from expert data rather than jumping straight into the learning phase, creating a more stable learning environment. They effectively addressed reinforcement overfitting by generating diverse self-play scenarios. While not explicitly discussed, AlphaGo raises important questions about the nature of human-computer interaction and creativity, especially as it challenges our understanding of intelligence by mastering a complex game like Go. My REU journey gave me a solid foundation in AI technology and strengthened my determination to contribute to this dynamic field.